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SPLUNK: EXPLORING SPL WITH TRY HACK ME

1.1 Introduction

In this module, I am introduced to Splunk, a powerful Security Information and Event Management (SIEM) platform used by analysts to search, visualize, and investigate large volumes of log data. I learned that Splunk's Search Processing Language (SPL) is essential for transforming raw data into meaningful insights.

Throughout this session, I explore how to build, filter, and refine search queries using SPL, enabling me to extract relevant information and identify potential security issues more efficiently. This module strengthens my ability to analyze data at scale and supports my growth as a security analyst by giving me practical, hands-on experience with a widely used industry tool.

2.1 Lab Access

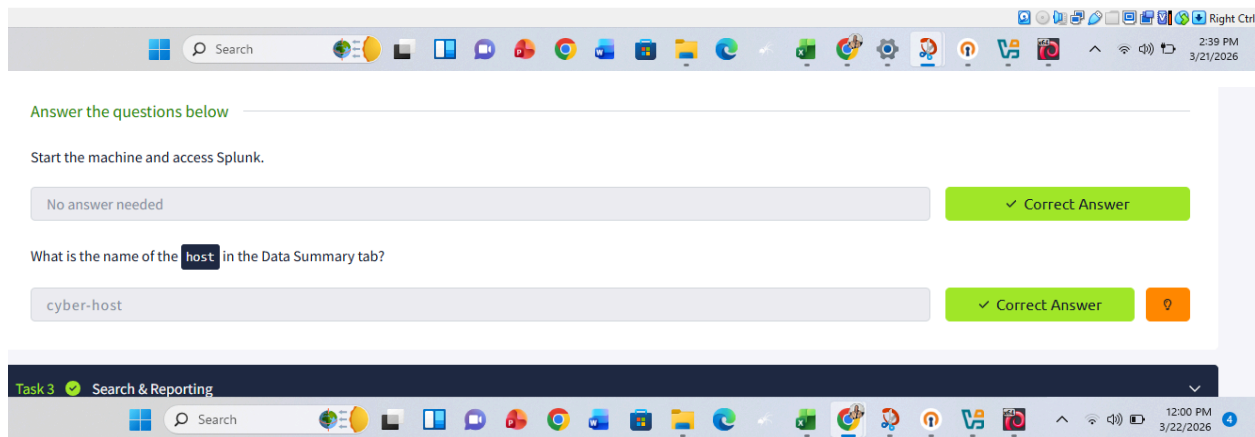
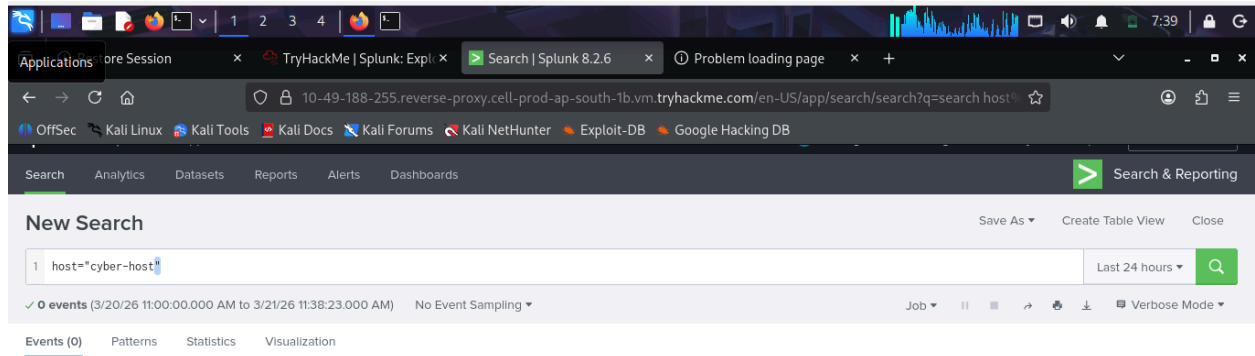
a) Start the machine and access Splunk.

<https://10-49-188-255.reverse-proxy.cell-prod-ap-south-1a.vm.tryhackme.com/>

This is how I accessed the Splunk platform

b) What is the name of the **host** in the Data Summary tab?

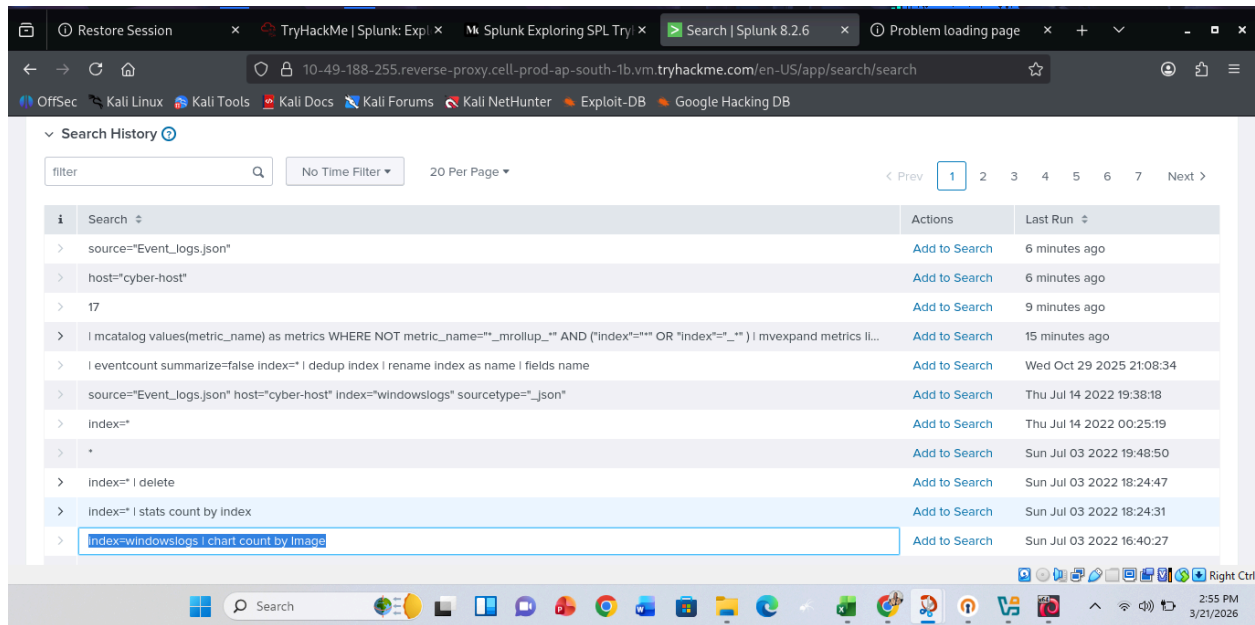
Answer: **cyber-host**



3.1 Search and Reporting

a) In the search History, what is the seventh search query in the list? (excluding any searches you have performed)

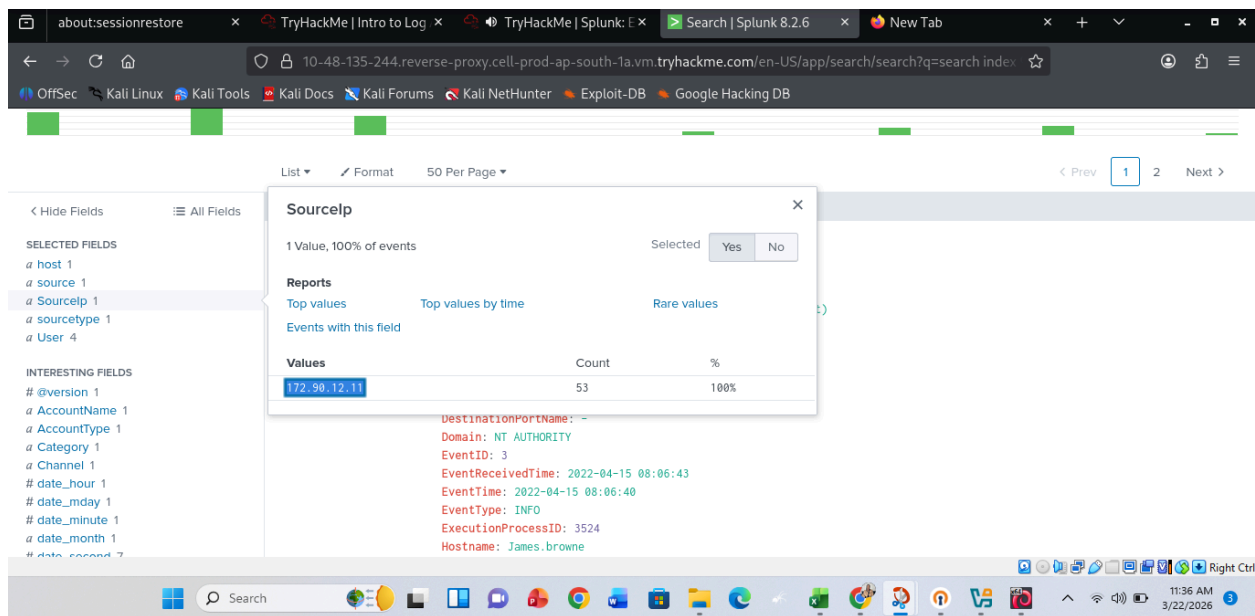
Answer: **index=windowslogs | chart count by Image**



b)After you submit your first query, look in the Fields sidebar.

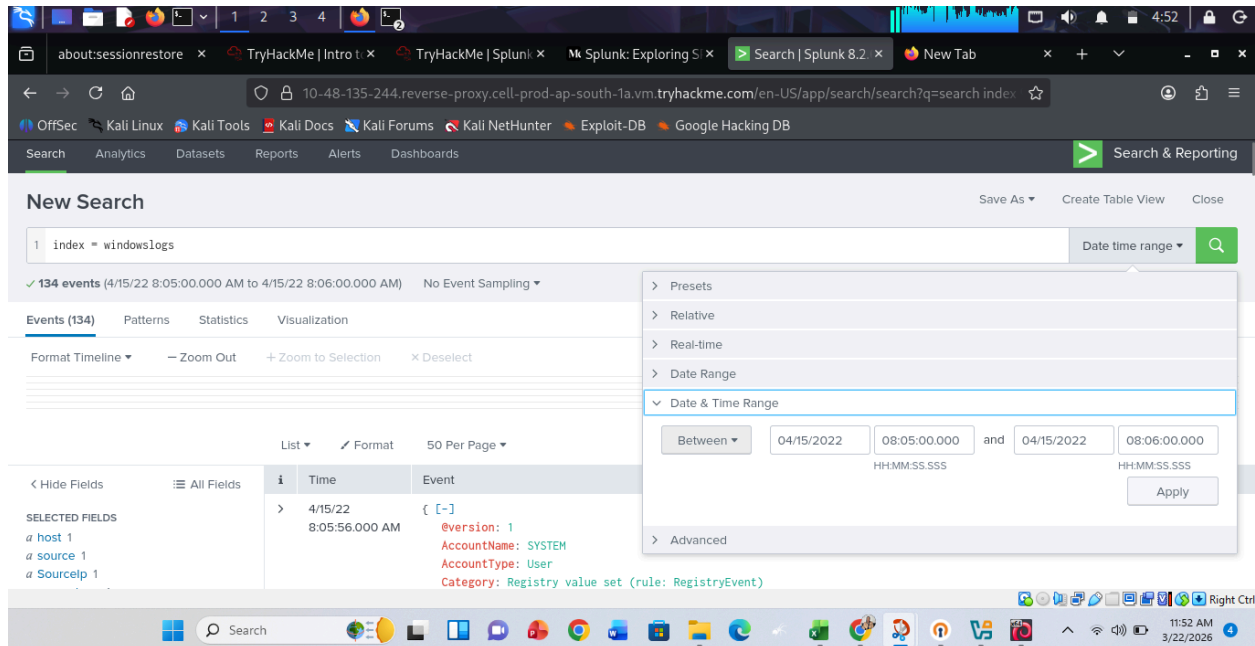
Which **SourceIP** has recorded the most amount of events?

Answer: **172.90.12.11**

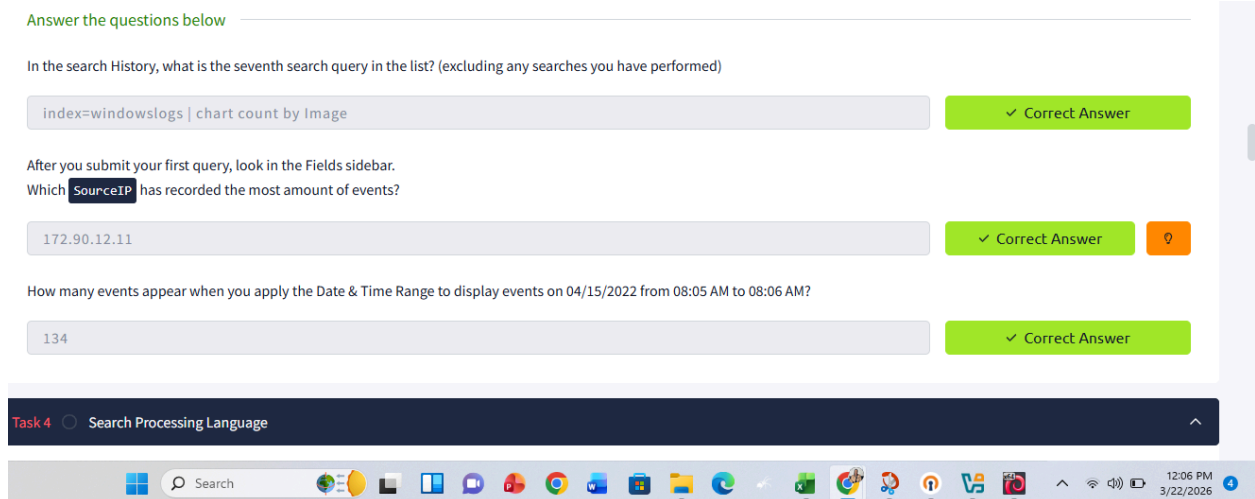


c)How many events appear when you apply the Date & Time Range to display events on 04/15/2022 from 08:05 AM to 08:06 AM?

Answer: **134**



Below is the screenshot to my answers



4.1 Search Processing Language

a) How many events in the `windowslogs` index have an `EventID` field value equal to `4624`?

Answer: 26

The screenshot shows the Splunk Enterprise interface. The search bar contains the query `index=windowslogs EventID=4624`. The results show 26 events. The event list is as follows:

Time	Event
4/15/22 8:06:38.000 AM	{ [-] @version: 1 ActivityID: {4F259F18-BCE1-0001-0E2B-5193808AD601}

b) How many events are observed with the `DestinationIp = 172.18.39.6` and `DestinationPort = 135`?

Answer: 4

The screenshot shows the Splunk Enterprise interface. The search bar contains the query `index=windowslogs DestinationIp = 172.18.39.6 AND DestinationPort = 135`. The results show 4 events. The event list is as follows:

Time	Event
4/15/22 8:06:02.000 AM	{ [-] @version: 1 AccountName: SYSTEM

c) Use the query `index = windowslogs Hostname = Salena.Adam DestinationIp = 172.18.38.5`. Which `SourceIp` returns the highest count?

Answer: 172.90.12.11

The screenshot shows a Splunk search interface with the following details:

- Search Query:** `index = windowslogs Hostname = Salena.Adam DestinationIp = 172.18.38.5`
- Results:** 19 events (before 3/22/26 12:00:01.000 PM). No Event Sampling.
- Visualization:** A bar chart showing event counts over time.
- Fields:**
 - SELECTED FIELDS:** `a host 1`, `a source 1`, `a SourceIp 2`, `a sourcetype 1`, `a User 2`.
 - INTERESTING FIELDS:** `# @version 1`, `a AccountName 1`, `a AccountType 1`.
- Modal Window:** A "SourceIp" modal window is open, showing a table of values:

Values	Count	%
172.90.12.11	17	89.474%
172.18.38.5	2	10.526%

d)How many events are returned when you search the term **cyber***?

Answer:12256

The screenshot shows a Splunk search interface with the following details:

- Search Query:** `cyber*`
- Results:** 12,256 events (before 3/22/26 12:02:00.000 PM). No Event Sampling.
- Visualization:** A bar chart showing event counts over time.
- Fields:**
 - SELECTED FIELDS:** `a host 1`, `a source 1`, `a SourceIp 7`, `a sourcetype 1`, `a User 4`.
 - INTERESTING FIELDS:** `# @version 1`, `a AccountName 4`, `a AccountType 2`.
- Event Details:** The first event is expanded, showing the following fields:

```

@version: 1
AccountName: SYSTEM
AccountType: User
CallTrace: C:\windows\SYSTEM32\ntdll.dll+9c534|C:\windows\SYSTEM32\psmserviceexhost.dll+222a3|C:\windows\SYSTEM32\psmserviceexhost.dll+1a172|C:\wind
Category: Process accessed (rule: ProcessAccess)
Channel: Microsoft-Windows-Sysmon/Operational
Domain: NT AUTHORITY
EventID: 10
EventReceivedTime: 2022-04-15 08:05:46
EventTime: 2022-04-15 08:05:44
EventTune: TNF0

```

4)Which operator is given the lowest priority in Splunk searches?

Answer: AND

Room progress (55%)

How many events in the `windowslogs` index have an `EventID` field value equal to `4624` ?

26 ✓ Correct Answer

How many events are observed with the `DestinationIp = 172.18.39.6` and `DestinationPort = 135` ?

4 ✓ Correct Answer

Use the query `index = windowslogs Hostname = Salena.Adam DestinationIp = 172.18.38.5`
Which `SourceIp` returns the highest count?

172.90.12.11 ✓ Correct Answer

How many events are returned when you search the term `cyber*` ?

12256 ✓ Correct Answer

Which operator is given the lowest priority in Splunk searches?

AND ✓ Correct Answer

5.1 Filtering Results

a) Use the `fields` command to highlight `Domain`, `SourceProcessId`, and `TargetProcessId`.
Which `SourceProcessId` has the highest value?

Answer: **9496**

TryHackMe | Splunk: Expl... Search | Splunk 8.2.6

10-48-139-167.reverse-proxy.cell-prod-ap-south-1a.vm.tryhackme.com/en-US/app/search/search?earliest=0&late...

OffSec Kali Linux Kali Tools Kali Docs Kali Forums Kali NetHunter Exploit-DB Google Hacking DB

Select Fields

Select All Within Filter Deselect All All fields Filter

+ Extract New Fields

Field	# of Values	Event Coverage	Type
Top values	Average over time	Events with this field	Min: 108
Top values by time	Maximum value over time		Max: 9992
Rare values	Minimum value time		Std Dev: 4058.9686320565593
9496	575	29.337%	
396	234	11.939%	
900	222	11.326%	
1616	165	8.418%	
876	137	6.99%	
0x924	133	6.786%	
2340	93	4.745%	

Account Domain: Cybertess
Login ID: 0x354

12:04 PM 3/23/2026

b) Try out this query `index = windowslogs | regex TargetObject = "Manager$"`
Which `TargetObject` field value contains the highest number of results?

Answer: **HKLM\SOFTWARE\Microsoft\SecurityManager**

The screenshot shows a Splunk search interface with a list of fields on the left and a 'TargetObject' field selected. The 'Values' table shows the following data:

Values	Count	%
HKLM\SOFTWARE\Microsoft\SecurityManager	64	46.043%
HKU\S-1-5-21-4020993649-1037605423-417876593-500	46	33.094%
\Software\Microsoft\Windows NT\CurrentVersion\HostActivityManager	28	20.144%
HKU\S-1-5-21-4020993649-1037605423-417876593-110	1	0.719%
4\Software\Microsoft\Windows NT\CurrentVersion\HostActivityManager	1	0.719%
HKLM\SOFTWARE\Microsoft\Windows\CurrentVersion\Explorer\SyncRootManager	1	0.719%

Below is a screenshot of my answers

The screenshot shows a quiz interface with two questions. The first question asks for the highest value of SourceProcessId, and the second question asks for the TargetObject field value with the highest number of results.

Question 1: Use the `fields` command to highlight `Domain`, `SourceProcessId`, and `TargetProcessId`. Which `SourceProcessId` has the highest value?

Answer: 9496

Question 2: Try out this query `index = windowslogs | regex TargetObject = "Manager$"`. Which `TargetObject` field value contains the highest number of results?

Answer: HKLM\SOFTWARE\Microsoft\SecurityManager

6.1 Structuring Search Results

a) Build a table that highlights the EventID, AccountName, and AccountType fields. Which AccountName appears first in your results?

Answer: **SYSTEM**

Search | Splunk 8.2.6

10-48-139-167.reverse-proxy.cell-prod-ap-south-1a.vm.tryhackme.com/en-US/app/search/search?earliest=0&latest=0

```

1 index=windowslogs
2 | fields EventID, AccountName, AccountType
3 | sort - AccountName

```

10,000 events (before 3/23/26 9:16:00 AM) No Event Sampling

AccountName

4 Values, 60.21% of events

Selected Yes No

Reports

Top values Top values by time Rare values

Events with this field

Values	Count	%
SYSTEM	5,937	98.605%
James	79	1.312%
NETWORK SERVICE	4	0.066%
LOCAL SERVICE	1	0.017%

b) Append the above query to include the reverse command. Which EventID appears first?
 Answer: **800**

Search | Splunk 8.2.6

10-48-139-167.reverse-proxy.cell-prod-ap-south-1a.vm.tryhackme.com/en-US/app/search/search?earliest=0&latest=0

```

1 index=windowslogs
2 | table EventID, AccountName, AccountType
3 | reverse

```

12,256 events (before 3/23/26 9:21:13.000 AM) No Event Sampling

Events (12,256) Patterns Statistics (12,256) Visualization

100 Per Page Format Preview

EventID	AccountName	AccountType
800	James	User
800	James	User
800	James	User
800	James	User
800	James	User

c) Use the query below to build a timeline of events. What password was given to the user Alberto?

index = windowslogs EventID = 1

| table _time ParentProcessId ProcessId ParentCommandLine CommandLine

| reverse

Answer: **paw0rd1**

New Search

```
1 index = windowslogs EventID = 1
2 | table _time ParentProcessId ProcessId ParentCommandLine CommandLine
3 | reverse
```

✓ 25 events (before 3/23/26 9:22:27.000 AM) No Event Sampling

Events (25) Patterns Statistics (25) Visualization

100 Per Page Format Preview

_time	ParentProcessId	ProcessId	ParentCommandLine	CommandLine
2022-04-15 08:06:01	9564	9428		"C:\windows\System32\Wbem\WMIC.exe" /node:WORKSTATION6 process call create "net user /add Alberto paw0rd1"
2022-04-15 08:06:02	900	3952	C:\windows\system32\svchost.exe -k DcomLaunch -p	C:\windows\system32\wbem\wmiprvse.exe -secured -Embedding
2022-04-15 08:06:02	3952	7768	C:\windows\system32\wbem\wmiprvse.exe -secured -Embedding	net user /add Alberto paw0rd1
2022-04-15 08:06:02	7768	7264	net user /add Alberto paw0rd1	\\?\C:\windows\system32\conhost.exe 0xffffffff -ForceV1

Below is the screenshot of my answers.

Room progress (80%)

Answer the questions below

Build a table that highlights the `EventID`, `AccountName`, and `AccountType` fields. Which `AccountName` appears first in your results?

SYSTEM ✓ Correct Answer

Append the above query to include the `reverse` command. Which `EventID` appears first?

800 ✓ Correct Answer

Use the query below to build a timeline of events. What password was given to the user `Alberto`?

```
index = windowslogs EventID = 1
| table _time ParentProcessId ProcessId ParentCommandLine CommandLine
| reverse
```

paw0rd1 ✓ Correct Answer

7.1 Transforming Commands

a) Use the `top` command to query the `Image` field.
Which `Image` field value has the most occurrences?
Answer: `C:\windows\system32\svchost.exe`

New Search

1 index=windowslogs | top limit=10 Image

✓ 12,256 events (before 3/23/26 10:01:00.000 AM) No Event Sampling

Events (12,256) Patterns **Statistics (10)** Visualization

100 Per Page Format Preview

Image	count	percent
C:\windows\system32\svchost.exe	1642	38.836329
C:\windows\system32\backgroundTaskHost.exe	547	12.937559
C:\Windows\System32\svchost.exe	426	10.075686
C:\windows\system32\taskhostw.exe	250	5.912961
C:\Windows\System32\BackgroundTransferHost.exe	210	4.966887

b) Try out the `iplocation` command with the `SourceIp` field.
Which **Region** do the IP addresses in your events originate from?

Answer: **California**

New Search

1 index = windowslogs | iplocation SourceIp | stats count by Region

✓ 12,256 events (before 3/23/26 9:59:27.000 AM) No Event Sampling

Events (12,256) Patterns **Statistics (1)** Visualization

100 Per Page Format Preview

Region	count
California	53

c) Try out this `lookup` query. Which **Image** field value has the highest **RiskScore**?
`index = windowslogs`

```
lookup image_riskScore Image OUTPUT RiskScore
stats count by Image RiskScore
sort - RiskScore
```

Answer: C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe

The screenshot shows a Splunk search interface with the following search query:

```
2 | lookup image_riskScore Image OUTPUT RiskScore
3 | stats count by Image RiskScore
4 | sort - RiskScore
```

The results table displays 12,256 events. The top results are as follows:

Image	RiskScore	count
C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe	10	15
C:\Windows\System32\lsass.exe	9	34
C:\Windows\system32\lsass.exe	9	35
C:\Windows\System32\wbem\WMIC.exe	7	67
C:\Windows\System32\net.exe	6	19
C:\Windows\System32\net1.exe	6	21
C:\Windows\system32\net1.exe	6	1
C:\Windows\ADWS\Microsoft.ActiveDirectory.WebServices.exe	5	2

Below is a screenshot of my answers.

The screenshot shows a TryHackMe room titled "splunkexploringspl" with the following tasks and answers:

- Task 1:** Use the `top` command to query the `image` field. Which `image` field value has the most occurrences?
 Answer: C:\windows\system32\svchost.exe ✓ Correct Answer
- Task 2:** Try out the `iplocation` command with the `sourceip` field. Which `region` do the IP addresses in your events originate from?
 Answer: California ✓ Correct Answer
- Task 3:** Try out this `lookup` query. Which `image` field value has the highest `riskScore`?

```
index = windows logs
| lookup image_riskScore Image OUTPUT RiskScore
| stats count by Image RiskScore
| sort - RiskScore
```

 Answer: C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe ✓ Correct Answer

Task 4: Conclusion

In this room, you learned how to use Splunk's Search Processing Language (SPL) to filter, structure, and transform raw log data into meaningful insights. You practiced using commands to build tables, charts, and visualizations to help uncover patterns and relationships between events.

Conclusion and Summary

Through this module, I have learned that SPL is a powerful tool for turning raw log data into meaningful insights. I now understand how to filter out noise using targeted searches, structure messy logs into clear and readable formats, and apply analytical commands to uncover patterns such as failed logins or suspicious activity. By using commands like `iplocation`, `top`, and `stats`, I can go beyond basic observation and actually interpret data in a way that supports decision-making. This experience has helped me transition from simply reading logs to actively investigating them, and I now feel more confident in my ability to identify issues, understand their impact, and ensure that security practices are being followed.

Below is a link and screenshot to the completion of the module

<https://tryhackme.com/room/splunkexploringspl/congratulations?step=records>

